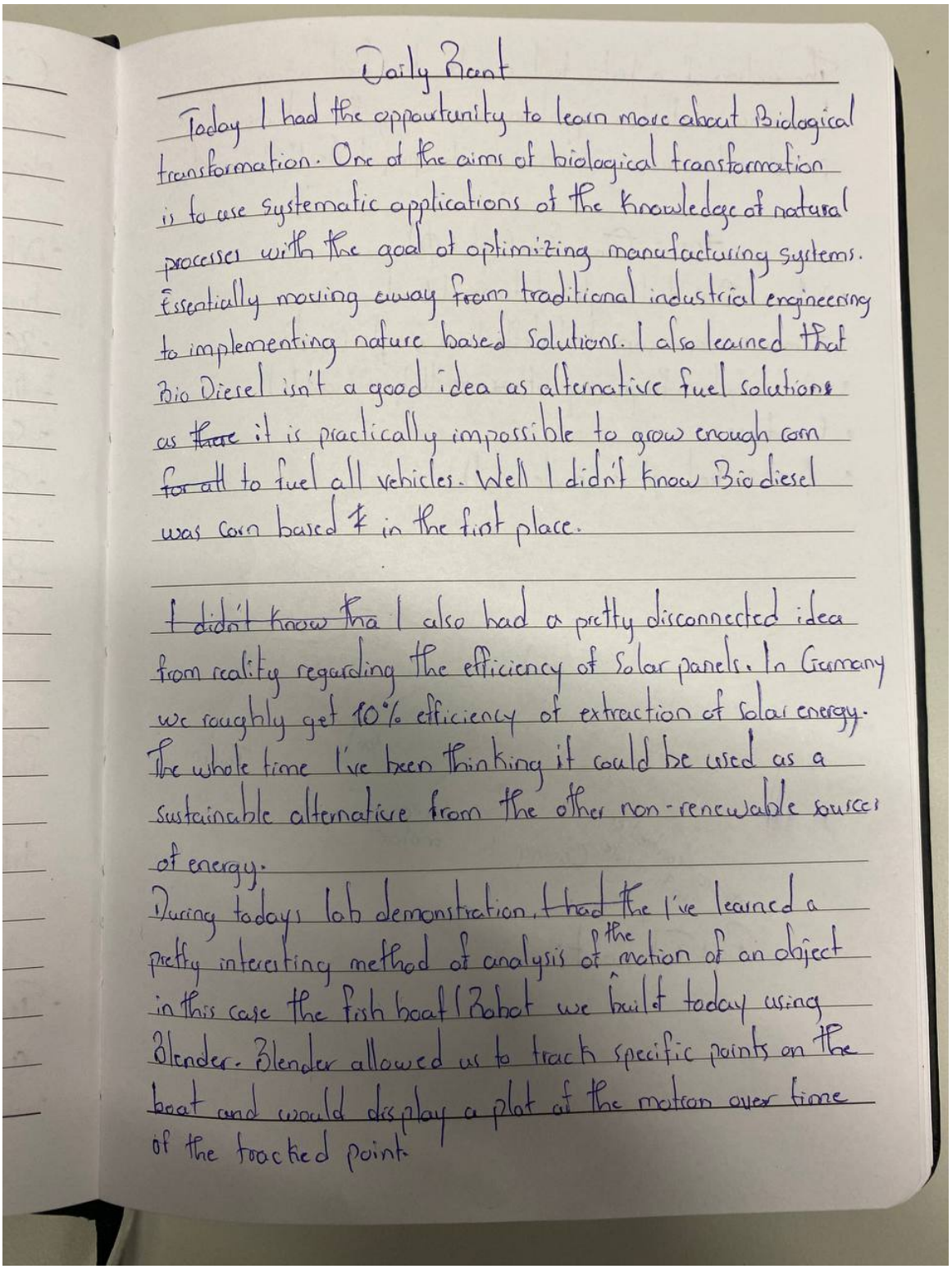
Biological Transformation

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Daily Rant

Today I had the opportunity to learn more about Biological transformation. One of the aims of biological transformation is to use systematic applications of the knowledge of natural processes with the goal of optimizing manufacturing systems. Essentially moving away from traditional industrial engineering to implementing nature based solutions. I also learned that Bio Diesel isn't a good idea as alternative fuel solutions as ~~there~~ it is practically impossible to grow enough corn ~~for all~~ to fuel all vehicles. Well I didn't know Biodiesel was corn based ☹ in the first place.

~~I didn't know tha~~ I also had a pretty disconnected idea from reality regarding the efficiency of Solar panels. In Germany we roughly get 10% efficiency of extraction of solar energy. The whole time I've been thinking it could be used as a sustainable alternative from the other non-renewable sources of energy.

During todays lab demonstration, ~~I had the~~ I've learned a pretty interesting method of analysis of the motion of an object in this case the fish boat/Robot we built today using Blender. Blender allowed us to track specific points on the boat and would display a plot of the motion over time of the tracked point.

The motion of a fish's tail can be described using a harmonic oscillation equation, assuming the tail moves periodically from side to side to propel itself forward.

$$\theta(t) = \theta_{max} \sin(\omega t + \phi)$$

- Where $\theta(t)$ tail angle at time t (rad)
- θ_{max} - maximum tail deflection angle (rad)
- ω - angular frequency of ~~of~~ oscillation (rad/s)
- t - time
- ϕ - initial phase angle (rad)

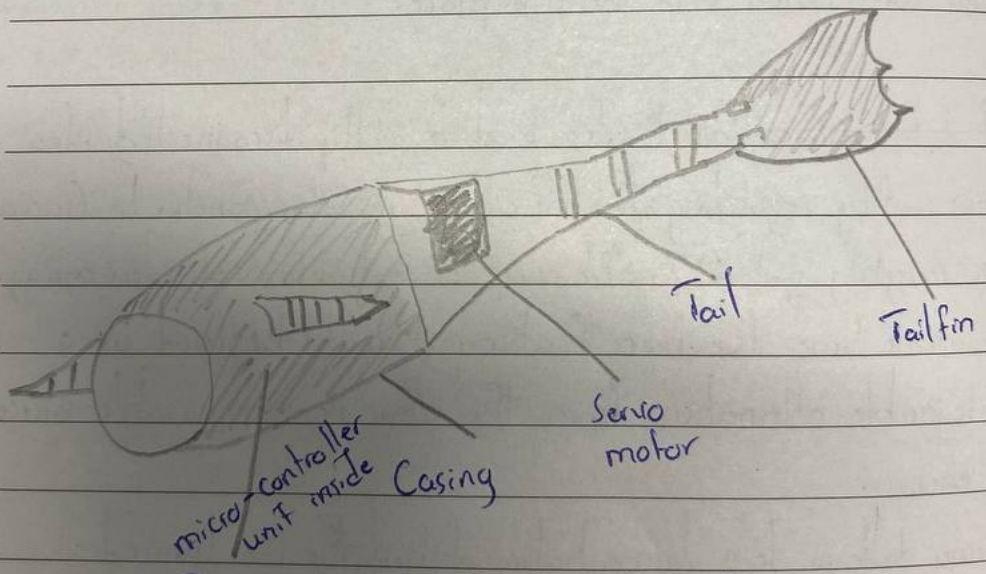


Fig: Fish boat: Motor Controlled through a micro-controller unit. Capable of ~~swimming~~ moving at the surface of water.